POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 5043A | Semester V | Theory | 4 Credits

Power Electronics & Automation

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Electronics Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 5043A as Power Electronics & Automation in Semester V for Electronics Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Kharagpur’s Industrial Automation and Control course and polytechnic power/automation curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Converter designs, PLC ladder logic, drive parameters and SCADA configurations must be based on approved industrial designs, certified automation software, current safety standards and competent professional review.

Verified source note: Verified sources support power devices, converters, industrial drives, PLC (ladder logic), and automation architecture. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Power Electronics & Automation studies the integration of high-power semiconductor devices with automated control systems for industrial applications. It develops the ability to analyze power converters (rectifiers, choppers, inverters), evaluate industrial drives and motor control, program Programmable Logic Controllers (PLC) using ladder logic, and understand SCADA and distributed control architectures while respecting the technical and safety boundaries of industrial automation.

> **Malayalam support:** □ handbook □□□□□□□□□□□□□ syllabus topic □□□□□□ □□□□□□□□□□□; □□□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□□ □□□□□□□□□ □□□□□□□□□□.

Prerequisite foundation

Basic electronics, electrical machines, digital electronics, control systems and engineering mathematics.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain the characteristics and applications of power semiconductor devices and converters in industrial systems.
2. Explain the operation and control of industrial motor drives using inverters and Variable Frequency Drives (VFD).
3. Develop PLC programs using ladder logic for sequence control and industrial automation tasks.
4. Explain the architecture and components of SCADA and distributed control systems for large-scale automation.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□□ exam-□ □□□□□□ □□□□□□□□ □□□□□□□ □□□□□□ □□□□□□□□□□. Define, explain, apply □□□□□□ action words □□□□□□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain the operation of power semiconductor devices and phase-controlled rectifiers.	Understand
CO2	Explain the principles of DC-DC choppers, inverters and industrial motor drives.	Understand
CO3	Develop ladder logic programs for PLC-based automation of industrial processes.	Apply
CO4	Explain the architecture of SCADA systems and industrial communication protocols.	Understand

Official CO-PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	**Title**	**Official major topics**	**Hours**	**CO**	**Handbook section**
1	Power Devices and Converters	Power semiconductor devices (SCR, MOSFET, IGBT); triggering and commutation; phase-controlled rectifiers (single-phase and three-phase); effect of loads; dual converters; industrial applications of rectifiers.	Approx. 16 hours	CO1	Module 1
2	Choppers, Inverters and Industrial Drives	DC-DC choppers (Buck, Boost); Single-phase and three-phase inverters; Pulse Width Modulation (PWM) techniques; Variable Frequency Drives (VFD); induction motor speed control; BLDC and Stepper motor drives.	Approx. 16 hours	CO2	Module 2
3	PLC and Ladder Logic Programming	PLC architecture and components; I/O modules; scan cycle; Relay Ladder Logic (RLL) fundamentals; timers, counters and registers; programming examples (motor starter, conveyor control); PLC selection and installation.	Approx. 14 hours	CO3	Module 3
4	Automation Systems and SCADA	Introduction to SCADA (Supervisory Control and Data Acquisition); SCADA architecture (MTU, RTU, HMI); industrial communication protocols (Modbus, Profibus); Distributed Control Systems (DCS); IoT in industrial automation.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Power Devices and Converters

Power semiconductor devices (SCR, MOSFET, IGBT); triggering and commutation; phase-controlled rectifiers (single-phase and three-phase); effect of loads; dual converters; industrial applications of rectifiers.

CO1 · Approx. 16 hours

Choppers, Inverters and Industrial Drives

DC-DC choppers (Buck, Boost); Single-phase and three-phase inverters; Pulse Width Modulation (PWM) techniques; Variable Frequency Drives (VFD); induction motor speed control; BLDC and Stepper motor drives.

CO2 · Approx. 16 hours

PLC and Ladder Logic Programming

PLC architecture and components; I/O modules; scan cycle; Relay Ladder Logic (RLL) fundamentals; timers, counters and registers; programming examples (motor starter, conveyor control); PLC selection and installation.

CO3 · Approx. 14 hours

Automation Systems and SCADA

Introduction to SCADA (Supervisory Control and Data Acquisition); SCADA architecture (MTU, RTU, HMI); industrial communication protocols (Modbus, Profibus); Distributed Control Systems (DCS); IoT in industrial automation.

CO4 · Approx. 14 hours

MODULE 1

Power Devices and Converters

Power semiconductor devices (SCR, MOSFET, IGBT); triggering and commutation; phase-controlled rectifiers (single-phase and three-phase); effect of loads; dual converters; industrial applications of rectifiers.

Approx. 16 hours **CO1** **Understand · Apply**

Learning goals

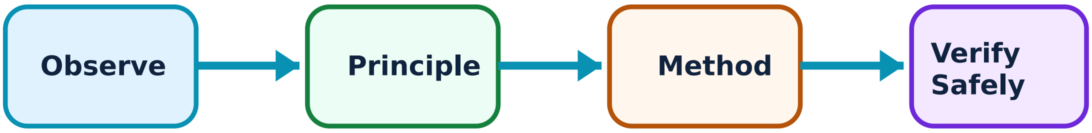
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Power Devices and Converters: identify the system, state its principle, follow the correct method and verify the result safely.

1. Power switches and rectifiers−

Power Electronics uses semiconductor switches to control electrical energy. The SCR is a key device for high-power rectifiers. Phase-controlled rectifiers convert AC to variable DC by adjusting the firing angle, used in industrial processes like electroplating and DC motor control.

Study and application method

Sketch the VI characteristics of an IGBT and compare it with a Power MOSFET; explain the operation of a three-phase full-bridge controlled rectifier.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the VI characteristics of an IGBT and compare it with a Power MOSFET; explain the operation of a three-phase full-bridge controlled rectifier.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: High-power converters involve dangerous voltages and currents. Do not operate converters without authorised safety enclosures and over-current protection.

2. Triggering and protection−

Triggering circuits provide the gate pulses for power devices. Protection involves snubbers (for dV/dt and dI/dt), fuses and heat sinks. Dual converters allow for four-quadrant operation, enabling both motoring and regenerative braking in DC drives.

Study and application method

Describe a UJT-based triggering circuit for an SCR; explain the role of a 'snubber circuit' in power device protection.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

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Common mistake / precaution: Incorrect triggering or inadequate cooling can lead to device failure or thermal runaway. Follow authorised thermal management and gate-drive specifications.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Choppers, Inverters and Industrial Drives

DC-DC choppers (Buck, Boost); Single-phase and three-phase inverters; Pulse Width Modulation (PWM) techniques; Variable Frequency Drives (VFD); induction motor speed control; BLDC and Stepper motor drives.

Approx. 16 hours | CO2 | Understand · Apply

Learning goals

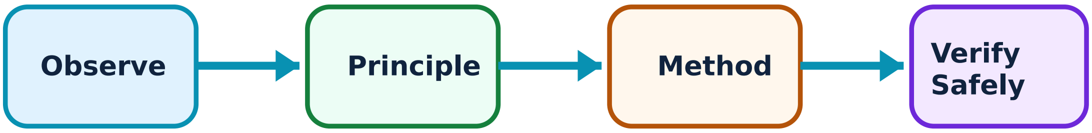
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Choppers, Inverters and Industrial Drives: identify the system, state its principle, follow the correct method and verify the result safely.

1. Converters and PWM−

Choppers provide variable DC from a fixed source. Inverters convert DC to AC, with PWM techniques controlling the output voltage and frequency. Sinusoidal PWM is critical for maintaining high power quality in industrial drives and renewable energy systems.

Study and application method

Explain the principle of a Boost converter conceptually; compare 180-degree and 120-degree conduction modes for a three-phase inverter.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the principle of a Boost converter conceptually; compare 180-degree and 120-degree conduction modes for a three-phase inverter.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Inverters produce high-frequency harmonics and EMI. Use authorised filters and shielding to prevent interference with control electronics and sensors.

2. Industrial motor drives−

Variable Frequency Drives (VFD) control the speed of induction motors by varying the supply frequency and voltage (V/f control). This significantly improves energy efficiency and process control. BLDC and stepper motor drives are used for precision motion in automation.

Study and application method

Sketch the block diagram of a VFD-based induction motor drive; explain the advantages of using VFDs for industrial pumping systems.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the block diagram of a VFD-based induction motor drive; explain the advantages of using VFDs for industrial pumping systems.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Motor drives involve complex parameter settings. Do not modify VFD parameters without authorised configuration and load-matching analysis.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

PLC and Ladder Logic Programming

PLC architecture and components; I/O modules; scan cycle; Relay Ladder Logic (RLL) fundamentals; timers, counters and registers; programming examples (motor starter, conveyor control); PLC selection and installation.

Approx. 14 hours | CO3 | Understand · Apply

Learning goals

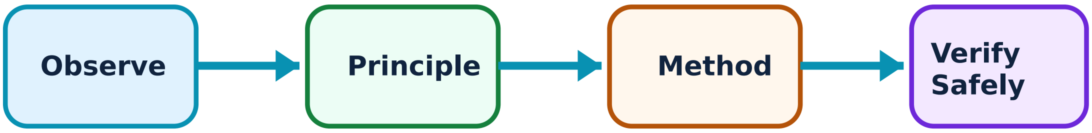
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for PLC and Ladder Logic Programming: identify the system, state its principle, follow the correct method and verify the result safely.

1. PLC hardware and scan cycle−

Programmable Logic Controllers (PLC) are rugged industrial computers for automation. The scan cycle involves reading inputs, executing the program and updating outputs. I/O modules interface the PLC with industrial sensors and actuators.

Study and application method

Sketch the architecture of a typical PLC; explain the three stages of a 'PLC Scan Cycle'.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the architecture of a typical PLC; explain the three stages of a 'PLC Scan Cycle'.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: PLC inputs and outputs have specific voltage and current ratings. Do not connect field devices without authorised wiring diagrams and isolation.

2. Ladder logic programming−

Ladder logic is a graphical programming language based on relay circuit diagrams. Contacts (NO/NC), coils, timers (On-delay/Off-delay) and counters are the basic elements used to implement complex sequential and combinational logic for automation.

Study and application method

Develop a ladder logic program for a simple 'Start-Stop' motor control with an overload interlock; explain the function of a 'Retentive Timer'.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Develop a ladder logic program for a simple 'Start-Stop' motor control with an overload interlock; explain the function of a 'Retentive Timer'.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Automation logic must include safety interlocks and emergency stops. All PLC programs must be thoroughly tested in a simulated environment before authorised deployment.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Automation Systems and SCADA

Introduction to SCADA (Supervisory Control and Data Acquisition); SCADA architecture (MTU, RTU, HMI); industrial communication protocols (Modbus, Profibus); Distributed Control Systems (DCS); IoT in industrial automation.

Approx. 14 hours | CO4 | Understand · Apply

Learning goals

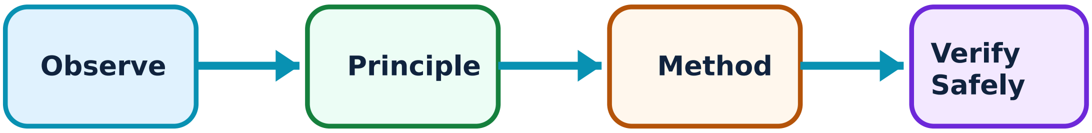
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Automation Systems and SCADA: identify the system, state its principle, follow the correct method and verify the result safely.

1. SCADA and HMI−

SCADA systems provide high-level supervision of industrial processes. Remote Terminal Units (RTU) collect data from the field, and the Master Terminal Unit (MTU) displays it on a Human Machine Interface (HMI) for operator control and data logging.

Study and application method

Sketch the basic architecture of a SCADA system; list the functions of an 'RTU' in a distributed automation network.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the basic architecture of a SCADA system; list the functions of an 'RTU' in a distributed automation network.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: SCADA systems are critical infrastructure. All network access must follow authorised cybersecurity protocols to prevent unauthorized control or data breaches.

2. Industrial communication and DCS−

Industrial protocols like Modbus and Profibus ensure reliable communication between automation devices. Distributed Control Systems (DCS) use multiple controllers distributed throughout a plant, providing high reliability and scalability for complex processes like oil refineries.

Study and application method

Compare SCADA and DCS in a conceptual table; explain the role of 'Fieldbus' in modern industrial communication.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare SCADA and DCS in a conceptual table; explain the role of 'Fieldbus' in modern industrial communication.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Communication failures in automation can lead to process upsets or safety hazards. Use authorised redundant communication paths and fail-safe logic for critical control loops.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

> For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

> For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

> For a comparison: use construction, working, result, advantage and limitation.

Recall 6

> For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Power Devices and Converters

Use the concept flow to revise observation, principle, method and verification.

Module 2: Choppers, Inverters and Industrial Drives

Use the concept flow to revise observation, principle, method and verification.

Module 3: PLC and Ladder Logic Programming

Use the concept flow to revise observation, principle, method and verification.

Module 4: Automation Systems and SCADA

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Compare SCR, MOSFET and IGBT in a conceptual table showing three industrial applications.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Sketch the output waveforms of a single-phase full-bridge inverter with PWM.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Develop a ladder logic program for a three-level water tank control system conceptually.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw the architecture of a SCADA system for a small-scale manufacturing plant.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify the sensors and actuators used in an automated conveyor or elevator system.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Power Devices and Converters

Explain Power switches and rectifiers and state one correct method, application or precaution.−

Power Electronics uses semiconductor switches to control electrical energy. The SCR is a key device for high-power rectifiers. Phase-controlled rectifiers convert AC to variable DC by adjusting the firing angle, used in industrial processes like electroplating and DC motor control.

Answer framework: Sketch the VI characteristics of an IGBT and compare it with a Power MOSFET; explain the operation of a three-phase full-bridge controlled rectifier.

Important point: High-power converters involve dangerous voltages and currents. Do not operate converters without authorised safety enclosures and over-current protection.

Explain Triggering and protection and state one correct method, application or precaution.−

Triggering circuits provide the gate pulses for power devices. Protection involves snubbers (for dV/dt and dI/dt), fuses and heat sinks. Dual converters allow for four-quadrant operation, enabling both motoring and regenerative braking in DC drives.

Answer framework: Describe a UJT-based triggering circuit for an SCR; explain the role of a 'snubber circuit' in power device protection.

Important point: Incorrect triggering or inadequate cooling can lead to device failure or thermal runaway. Follow authorised thermal management and gate-drive specifications.

Module 2 — Choppers, Inverters and Industrial Drives

Explain Converters and PWM and state one correct method, application or precaution.−

Choppers provide variable DC from a fixed source. Inverters convert DC to AC, with PWM techniques controlling the output voltage and frequency. Sinusoidal PWM is critical for maintaining high power quality in industrial drives and renewable energy systems.

Answer framework: Explain the principle of a Boost converter conceptually; compare 180-degree and 120-degree conduction modes for a three-phase inverter.

Important point: Inverters produce high-frequency harmonics and EMI. Use authorised filters and shielding to prevent interference with control electronics and sensors.

Explain Industrial motor drives and state one correct method, application or precaution.−

Variable Frequency Drives (VFD) control the speed of induction motors by varying the supply frequency and voltage (V/f control). This significantly improves energy efficiency and process control. BLDC and stepper motor drives are used for precision motion in automation.

Answer framework: Sketch the block diagram of a VFD-based induction motor drive; explain the advantages of using VFDs for industrial pumping systems.

Important point: Motor drives involve complex parameter settings. Do not modify VFD parameters without authorised configuration and load-matching analysis.

Module 3 — PLC and Ladder Logic Programming

Explain PLC hardware and scan cycle and state one correct method, application or precaution.−

Programmable Logic Controllers (PLC) are rugged industrial computers for automation. The scan cycle involves reading inputs, executing the program and updating outputs. I/O modules interface the PLC with industrial sensors and actuators.

Answer framework: Sketch the architecture of a typical PLC; explain the three stages of a 'PLC Scan Cycle'.

Important point: PLC inputs and outputs have specific voltage and current ratings. Do not connect field devices without authorised wiring diagrams and isolation.

Explain Ladder logic programming and state one correct method, application or precaution.−

Ladder logic is a graphical programming language based on relay circuit diagrams. Contacts (NO/NC), coils, timers (On-delay/Off-delay) and counters are the basic elements used to implement complex sequential and combinational logic for automation.

Answer framework: Develop a ladder logic program for a simple 'Start-Stop' motor control with an overload interlock; explain the function of a 'Retentive Timer'.

Important point: Automation logic must include safety interlocks and emergency stops. All PLC programs must be thoroughly tested in a simulated environment before authorised deployment.

Module 4 — Automation Systems and SCADA

Explain SCADA and HMI and state one correct method, application or precaution.−

SCADA systems provide high-level supervision of industrial processes. Remote Terminal Units (RTU) collect data from the field, and the Master Terminal Unit (MTU) displays it on a Human Machine Interface (HMI) for operator control and data logging.

Answer framework: Sketch the basic architecture of a SCADA system; list the functions of an 'RTU' in a distributed automation network.

Important point: SCADA systems are critical infrastructure. All network access must follow authorised cybersecurity protocols to prevent unauthorized control or data breaches.

Explain Industrial communication and DCS and state one correct method, application or precaution.−

Industrial protocols like Modbus and Profibus ensure reliable communication between automation devices. Distributed Control Systems (DCS) use multiple controllers distributed throughout a plant, providing high reliability and scalability for complex processes like oil refineries.

Answer framework: Compare SCADA and DCS in a conceptual table; explain the role of 'Fieldbus' in modern industrial communication.

Important point: Communication failures in automation can lead to process upsets or safety hazards. Use authorised redundant communication paths and fail-safe logic for critical control loops.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Power Devices and Converters.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Choppers, Inverters and Industrial Drives.

4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: PLC and Ladder Logic Programming.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Automation Systems and SCADA.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Power semiconductor devices (SCR, MOSFET, IGBT); triggering and commutation; phase-controlled rectifiers (single-phase and three-phase); effect of loads; dual converters; industrial applications of rectifiers..
2. Develop a complete answer or practical plan for: DC-DC choppers (Buck, Boost); Single-phase and three-phase inverters; Pulse Width Modulation (PWM) techniques; Variable Frequency Drives (VFD); induction motor speed control; BLDC and Stepper motor drives..
3. Develop a complete answer or practical plan for: PLC architecture and components; I/O modules; scan cycle; Relay Ladder Logic (RLL) fundamentals; timers, counters and registers; programming examples (motor starter, conveyor control); PLC selection and installation..
4. Develop a complete answer or practical plan for: Introduction to SCADA (Supervisory Control and Data Acquisition); SCADA architecture (MTU, RTU, HMI); industrial communication protocols (Modbus, Profibus); Distributed Control Systems (DCS); IoT in industrial automation..

LAST-MILE REVISION

Quick Revision

Module 1: Power Devices and Converters

Power semiconductor devices (SCR, MOSFET, IGBT); triggering and commutation; phase-controlled rectifiers (single-phase and three-phase); effect of loads; dual converters; industrial applications of rectifiers.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Choppers, Inverters and Industrial Drives

DC-DC choppers (Buck, Boost); Single-phase and three-phase inverters; Pulse Width Modulation (PWM) techniques; Variable Frequency Drives (VFD); induction motor speed control; BLDC and Stepper motor drives.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: PLC and Ladder Logic Programming

PLC architecture and components; I/O modules; scan cycle; Relay Ladder Logic (RLL) fundamentals; timers, counters and registers; programming examples (motor starter, conveyor control); PLC selection and installation.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Automation Systems and SCADA

Introduction to SCADA (Supervisory Control and Data Acquisition); SCADA architecture (MTU, RTU, HMI); industrial communication protocols (Modbus, Profibus); Distributed Control Systems (DCS); IoT in industrial automation.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus

Term / topic	One-line meaning
Power switches and rectifiers	Power Electronics uses semiconductor switches to control electrical energy.
Triggering and protection	Triggering circuits provide the gate pulses for power devices.
Converters and PWM	Choppers provide variable DC from a fixed source.
Industrial motor drives	Variable Frequency Drives (VFD) control the speed of induction motors by varying the supply frequency and voltage (V/f control).
PLC hardware and scan cycle	Programmable Logic Controllers (PLC) are rugged industrial computers for automation.
Ladder logic programming	Ladder logic is a graphical programming language based on relay circuit diagrams.
SCADA and HMI	SCADA systems provide high-level supervision of industrial processes.
Industrial communication and DCS	Industrial protocols like Modbus and Profibus ensure reliable communication between automation devices.

LEARNING RESOURCES

References

Recommended power-automation references

1. S. Mukhopadhyay, S. Sen and A. K. Deb, Industrial Instrumentation, Control and Automation.
2. John W. Webb and Ronald A. Reis, Programmable Logic Controllers: Principles and Applications.
3. M. H. Rashid, Power Electronics: Circuits, Devices and Applications.
4. Stuart A. Boyer, SCADA: Supervisory Control and Data Acquisition.

Approved public power-automation sources

- https://nptel.ac.in/courses/108105062

- https://nptel.ac.in/courses/108108077

These sources provide the verified study basis while official Course 5043A detail is unavailable; they do not replace official industrial designs, authorised automation software, certified safety codes or authorised production plans.